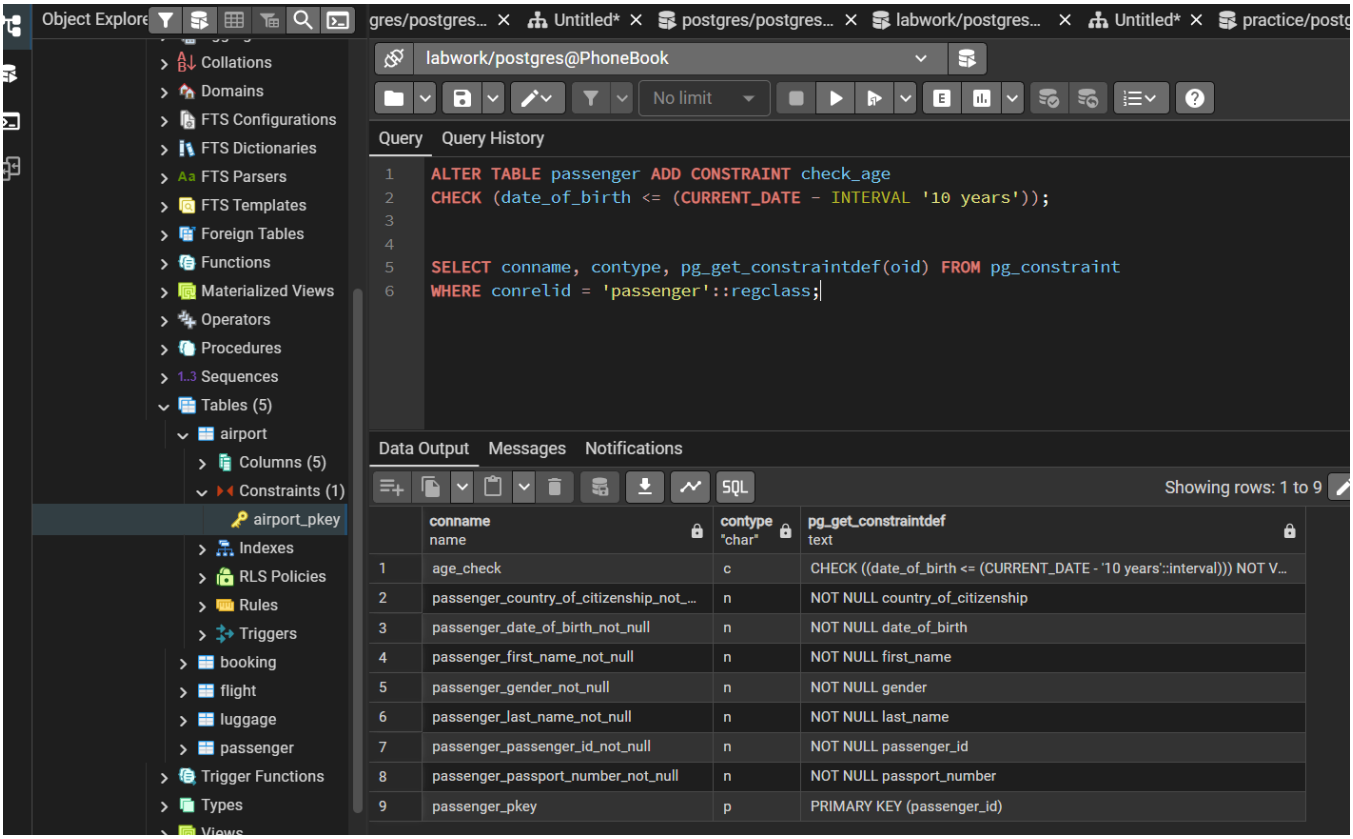
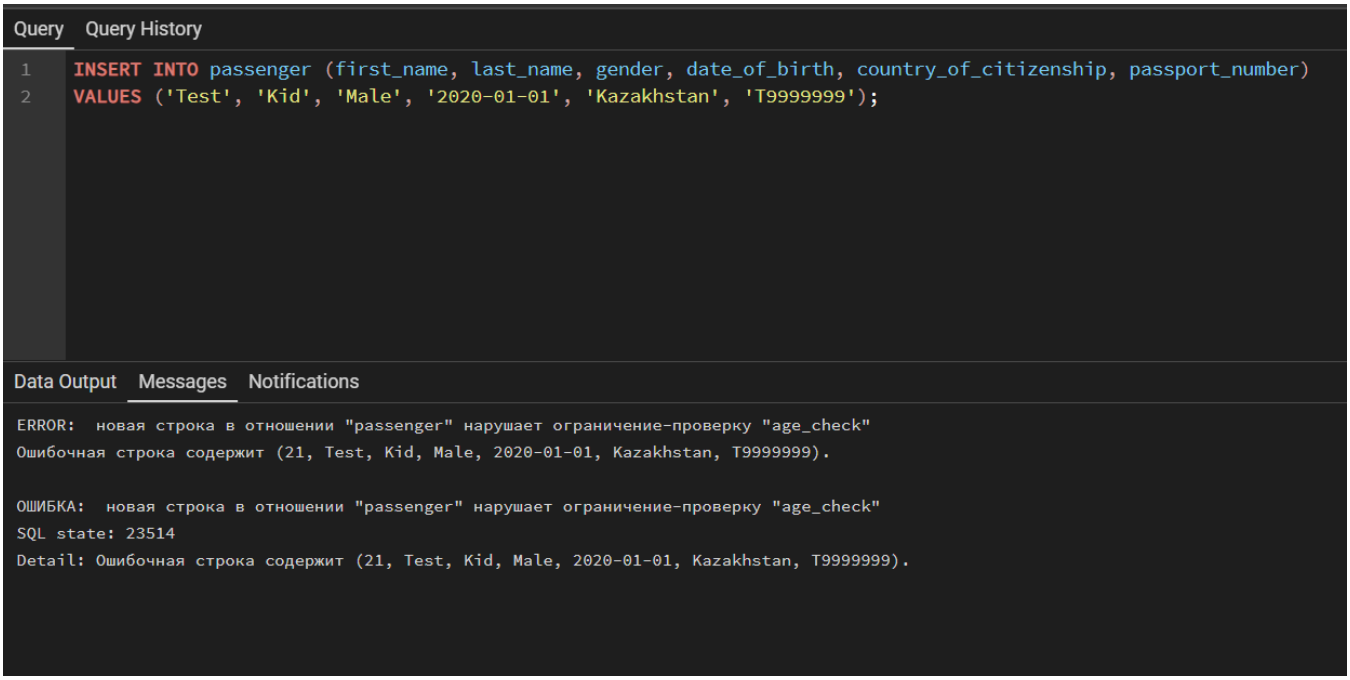
DATABASE
labaratory work 5

Task 1: Add a CHECK constraint to passenger table to provide that passengers must be at least 10 years old.

ALTER TABLE passenger ADD CONSTRAINT age_check
CHECK (date_of_birth <= (CURRENT_DATE - INTERVAL '10 years'))





Task 2: Add a CHECK constraint to accept values in booking price not more than 50000tg and less than 0tg.

ALTER TABLE booking ADD CONSTRAINT book_price_check
CHECK (booking_price >= 0 AND booking_price <= 50000)

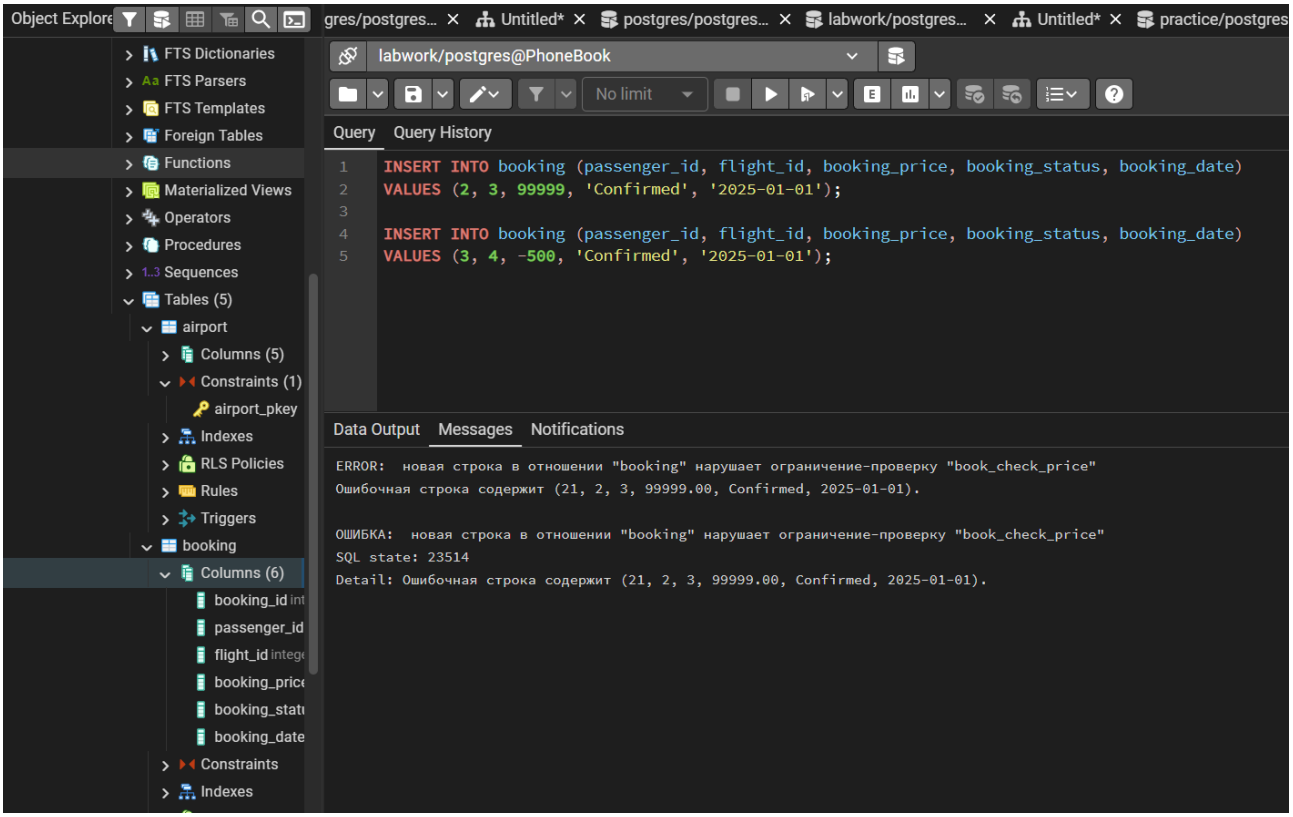
Query | Query History

```
1  ALTER TABLE booking ADD CONSTRAINT book_check_price
2  CHECK(booking_price <= 50000 AND booking_price >= 0) NOT VALID;
3
4  SELECT conname, contype, pg_get_constraintdef(oid) FROM pg_constraint
5  WHERE conrelid = 'booking'::regclass;
```

Data Output | Messages | Notifications

Showing rows: 1 to 8

	conname name	contype "char"	pg_get_constraintdef text
1	book_check_price	c	CHECK (((booking_price <= (50000)::numeric) AND (booking_price >= (0)::numeric))) NOT V...
2	booking_booking_id_not_null	n	NOT NULL booking_id
3	booking_booking_price_not_n...	n	NOT NULL booking_price
4	booking_flight_id_fkey	f	FOREIGN KEY (flight_id) REFERENCES flight(flight_id)
5	booking_flight_id_not_null	n	NOT NULL flight_id
6	booking_passenger_id_fkey	f	FOREIGN KEY (passenger_id) REFERENCES passenger(passenger_id)
7	booking_passenger_id_not_n...	n	NOT NULL passenger_id
8	booking_pkey	p	PRIMARY KEY (booking_id)



Task 3: Add a CHECK constraint to accept the luggage weight between 1 and 23 kg.

ALTER TABLE luggage ADD CONSTRAINT check_weight
CHECK (luggage_weight BETWEEN 1 AND 23)

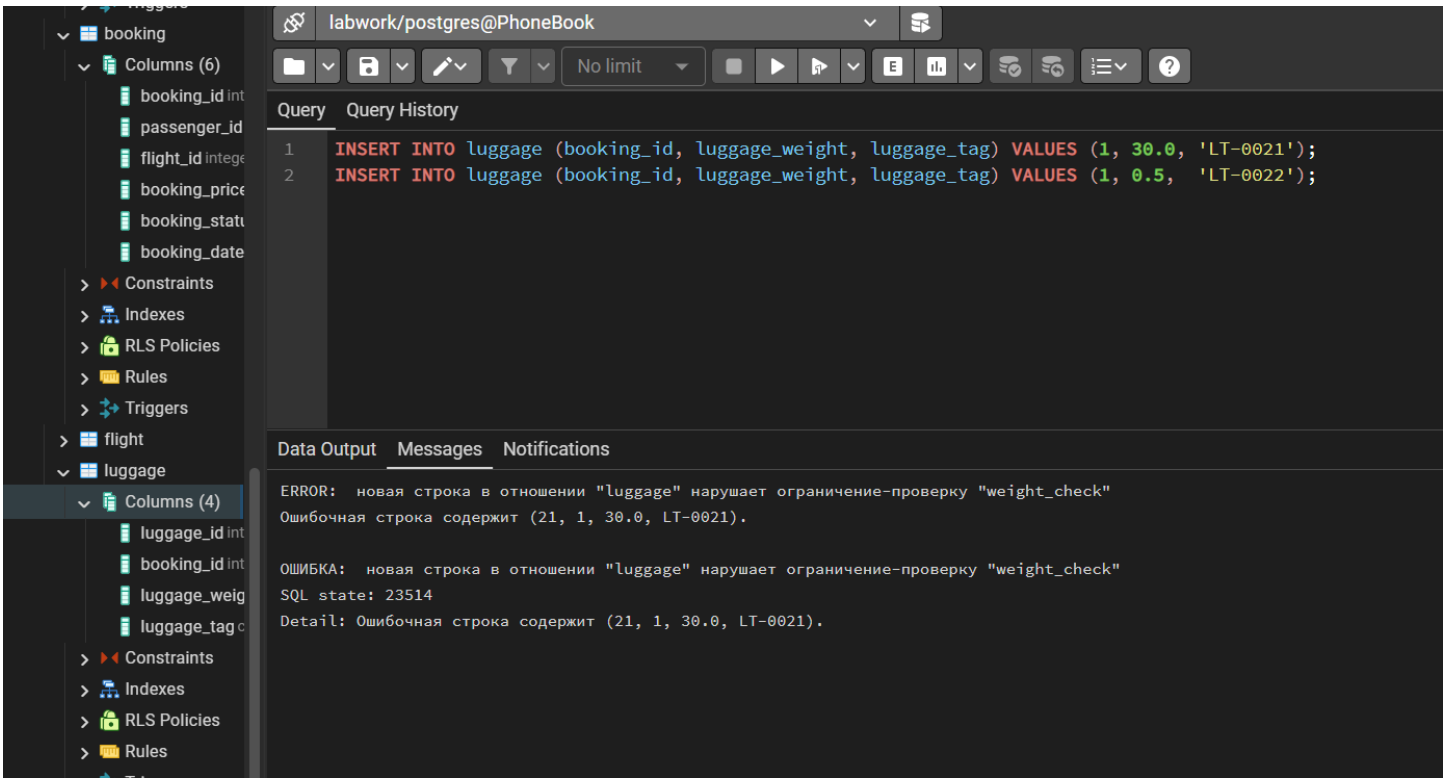
Query | Query History

```
1  ALTER TABLE luggage ADD CONSTRAINT weight_check
2  CHECK(luggage_weight > 1 AND luggage_weight < 23) NOT VALID;
3
4  SELECT conname, contype, pg_get_constraintdef(oid) FROM pg_constraint
5  WHERE conrelid = 'luggage'::regclass;
```

Data Output | Messages | Notifications

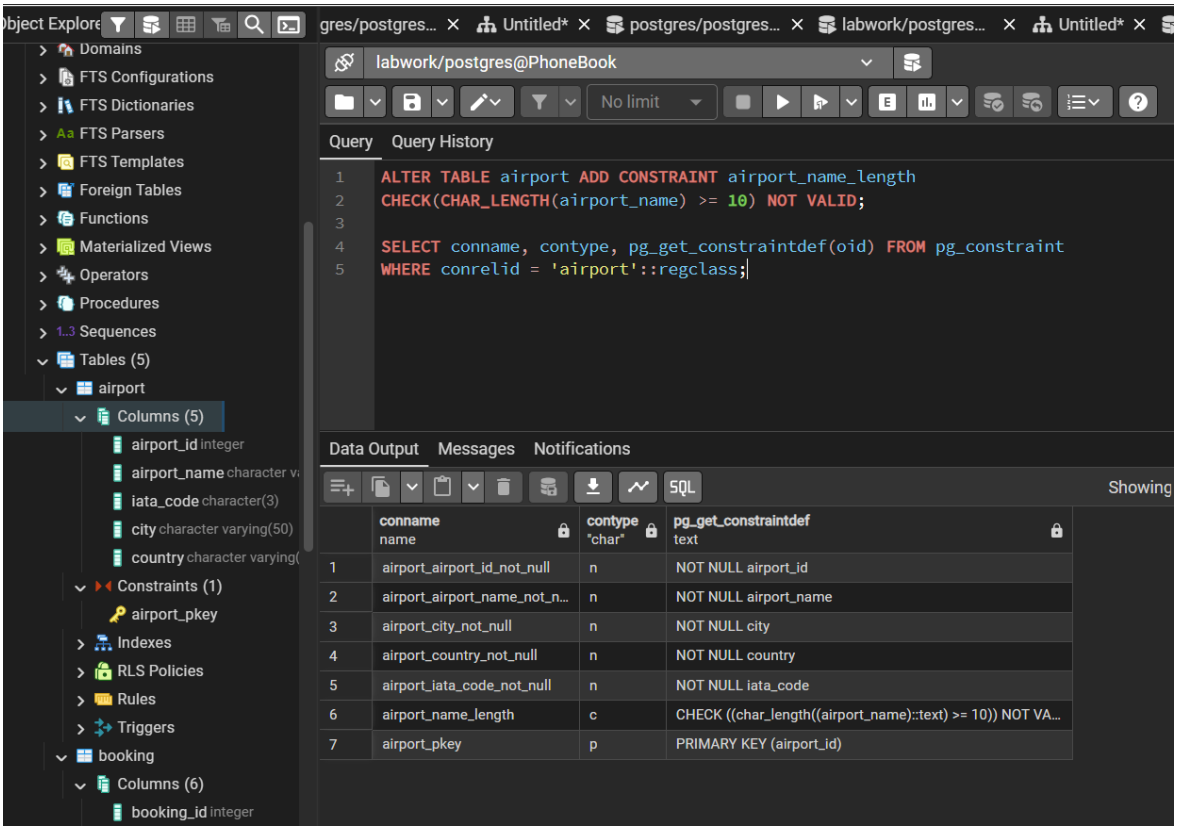
Showing rows: 1 to 7

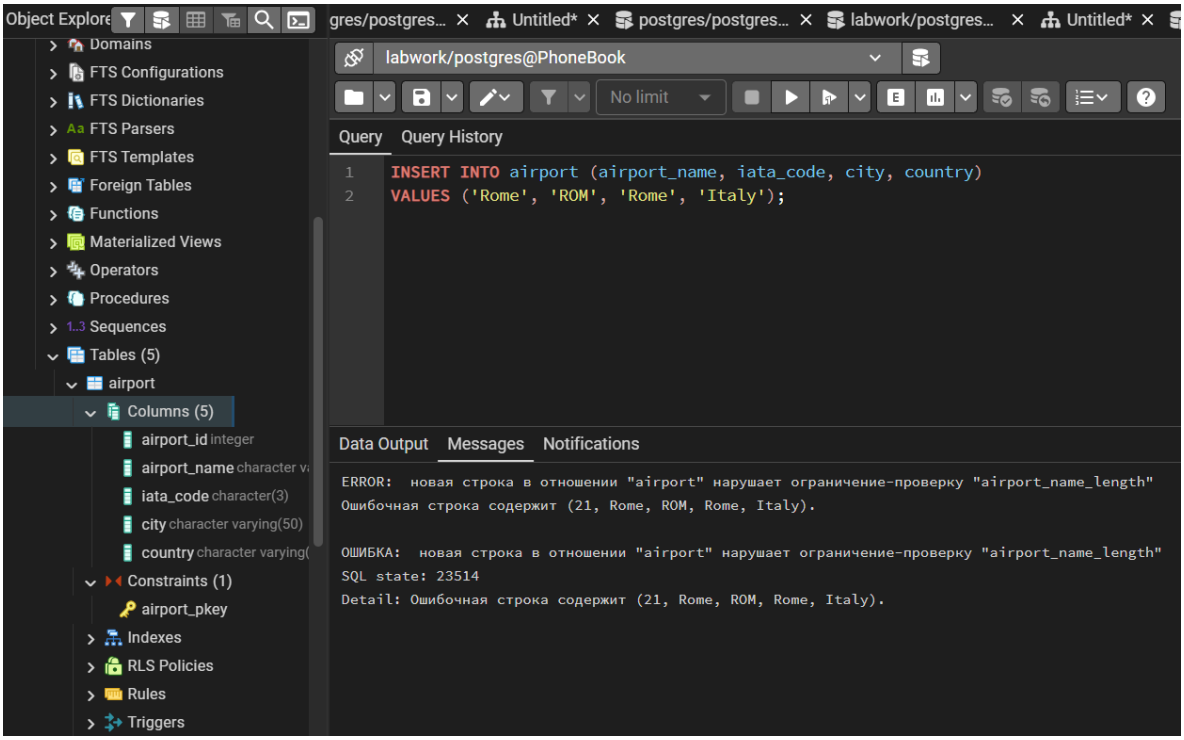
	conname name	contype "char"	pg_get_constraintdef text
1	luggage_booking_id_fkey	f	FOREIGN KEY (booking_id) REFERENCES booking(booking_id)
2	luggage_booking_id_not_null	n	NOT NULL booking_id
3	luggage_luggage_id_not_null	n	NOT NULL luggage_id
4	luggage_luggage_tag_not_null	n	NOT NULL luggage_tag
5	luggage_luggage_weight_not_n...	n	NOT NULL luggage_weight
6	luggage_pkey	p	PRIMARY KEY (luggage_id)
7	weight_check	c	CHECK (((luggage_weight > (1)::numeric) AND (luggage_weight < (23)::numeric))) NOT V...



Task 4: Add a CHECK constraint to ensure that all values in airport_name must have at least 10 characters.

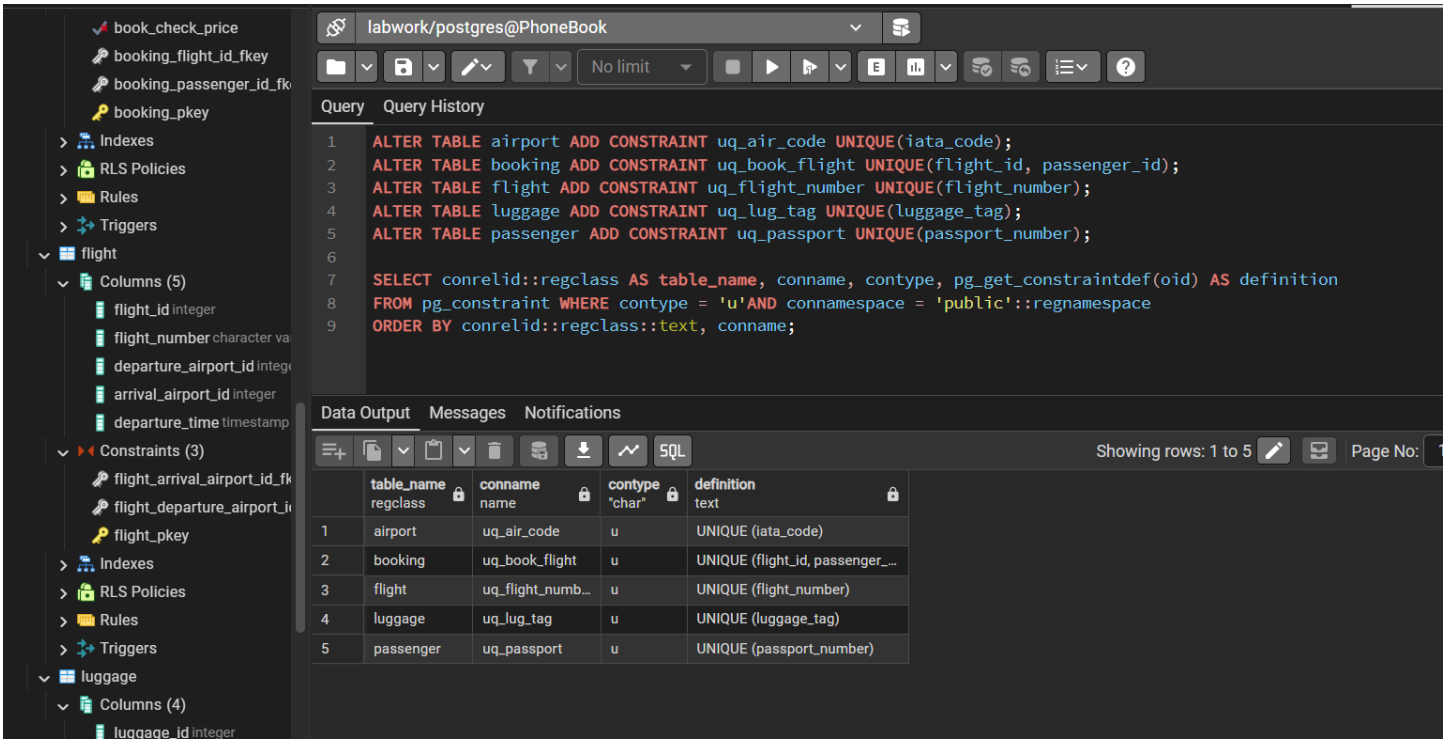
ALTER TABLE airport ADD CONSTRAINT chk_airport_name_len
CHECK (char_length(airport_name) >= 10)

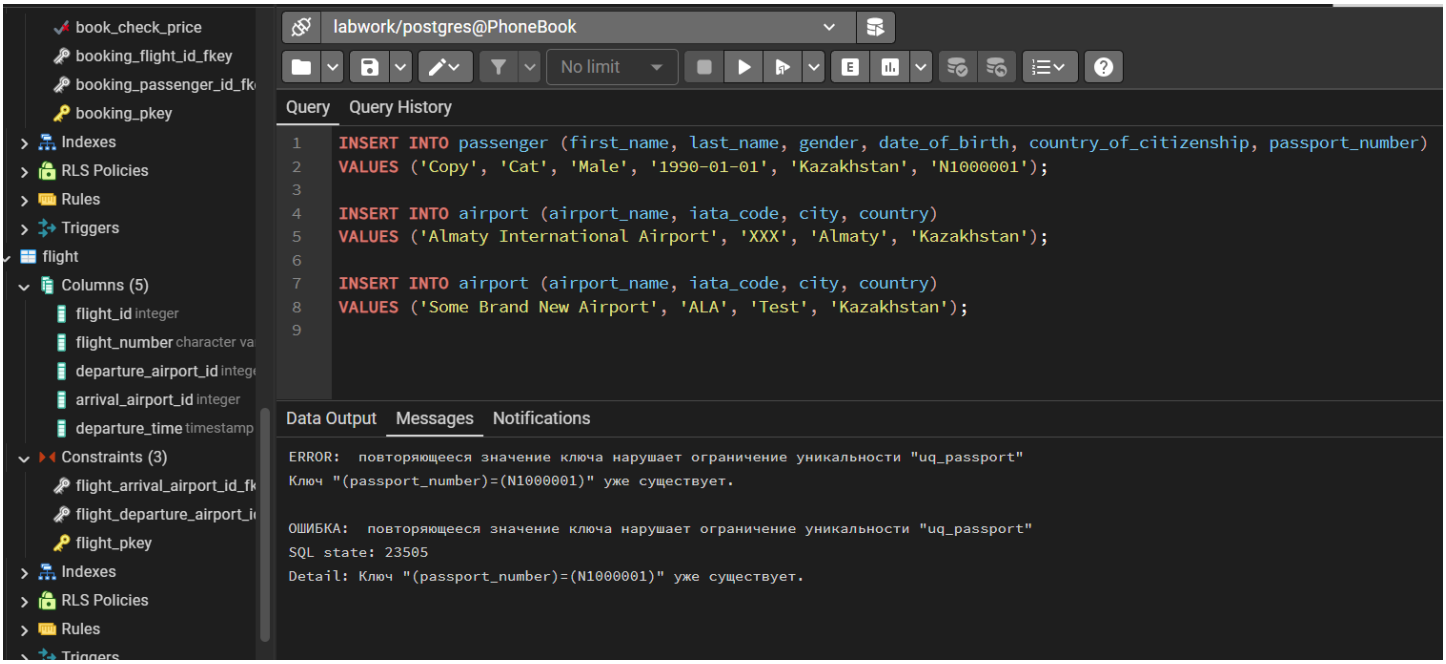




Task 5: Add UNIQUE constraint to some columns in each table in database.

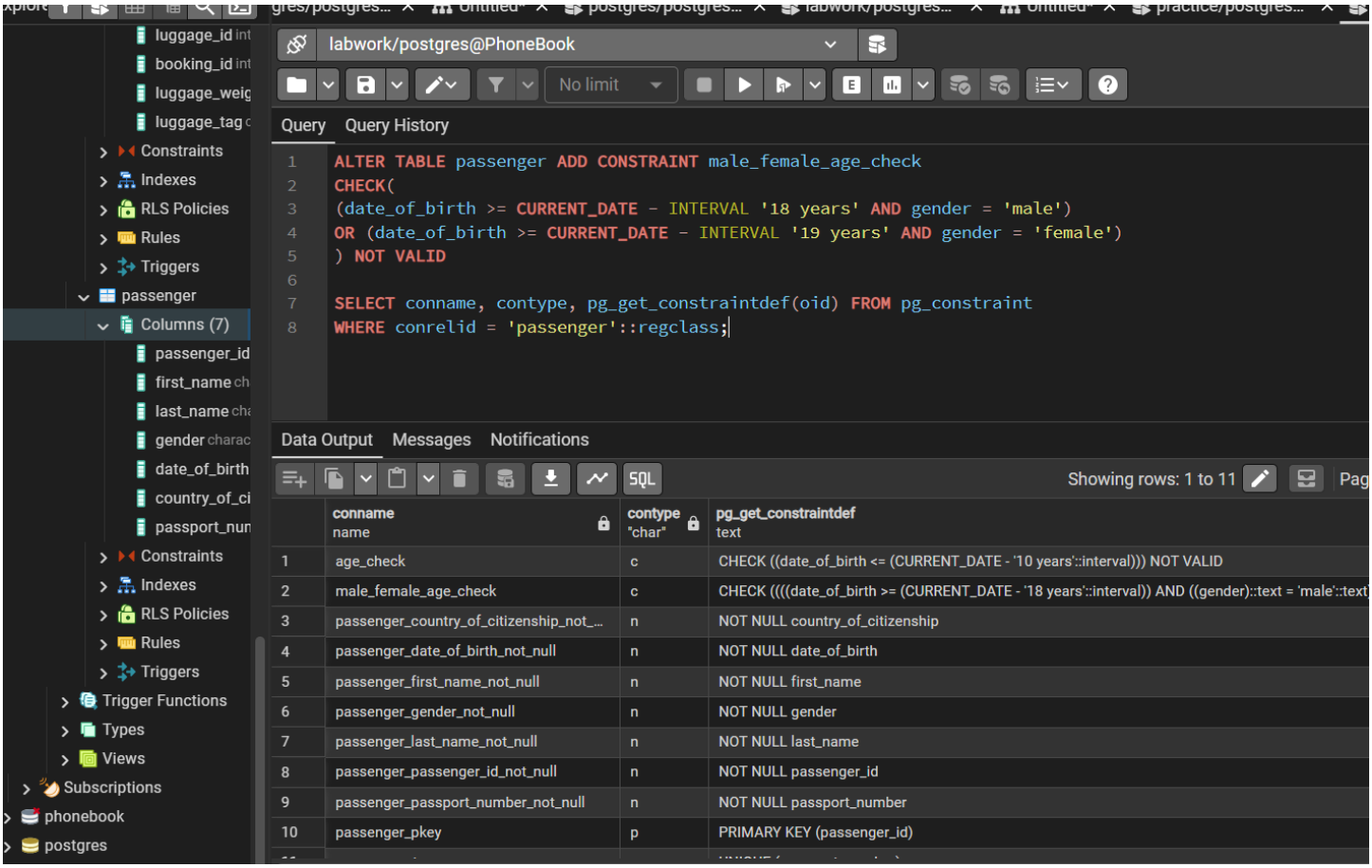
ALTER TABLE passenger ADD CONSTRAINT uq_passport UNIQUE (passport_number);
ALTER TABLE airport ADD CONSTRAINT uq_airport_name UNIQUE (airport_name);
ALTER TABLE airport ADD CONSTRAINT uq_airport_iata UNIQUE (iata_code);
ALTER TABLE luggage ADD CONSTRAINT uq_luggage_tag UNIQUE (luggage_tag);
ALTER TABLE booking ADD CONSTRAINT uq_booking_pass_flight UNIQUE (passenger_id, flight_id);

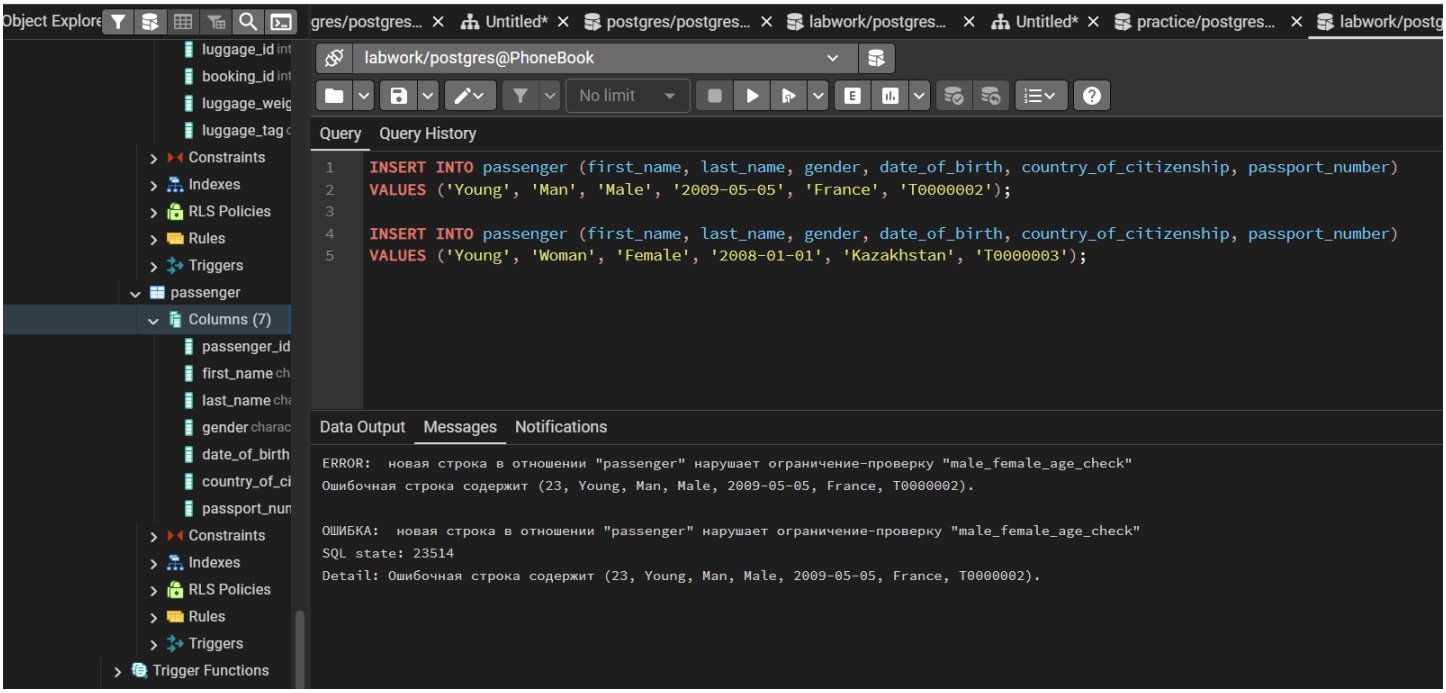




Task 6: Add a CHECK constraint to ensure that male passengers must be at least 18 years old and female passengers must be 19 years old.

ALTER TABLE passenger ADD CONSTRAINT chk_gender_min_age
CHECK (
(gender = 'Male' AND date_of_birth <= CURRENT_DATE - INTERVAL '18 years') OR
(gender = 'Female' AND date_of_birth <= CURRENT_DATE - INTERVAL '19 years'))



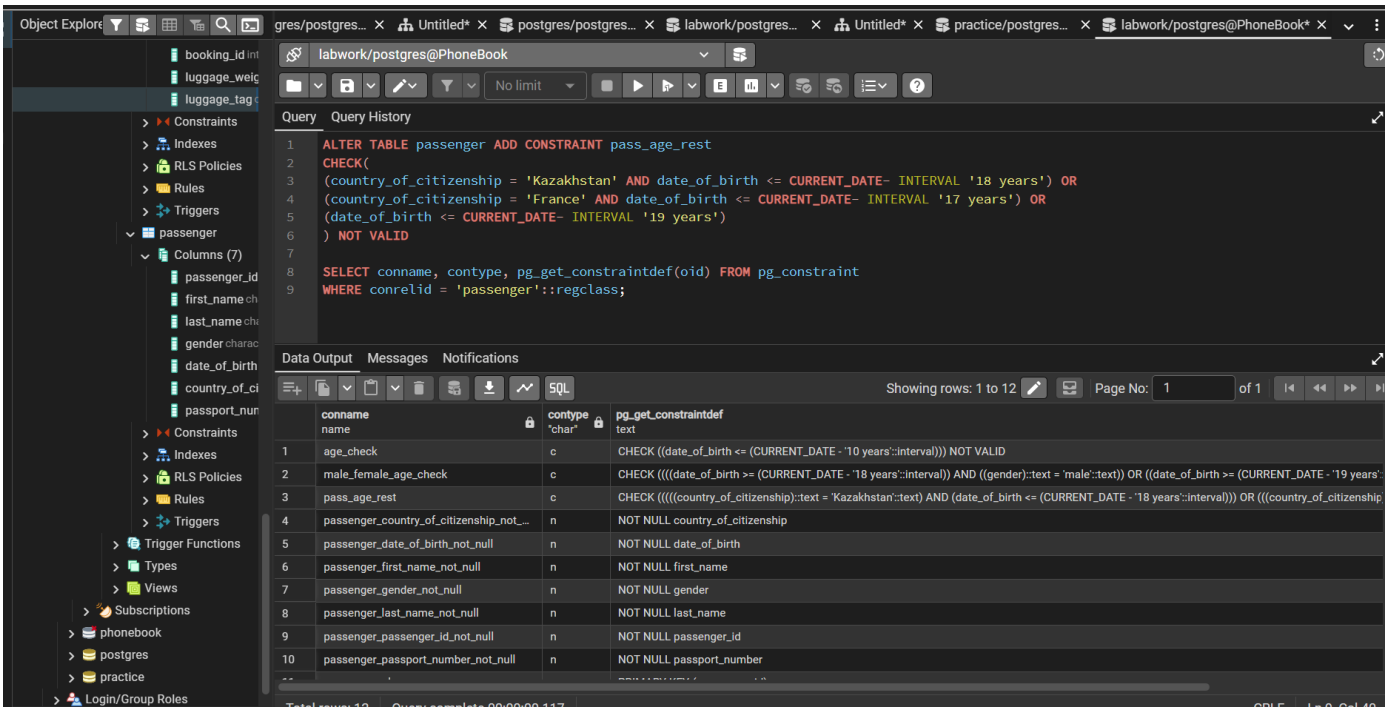


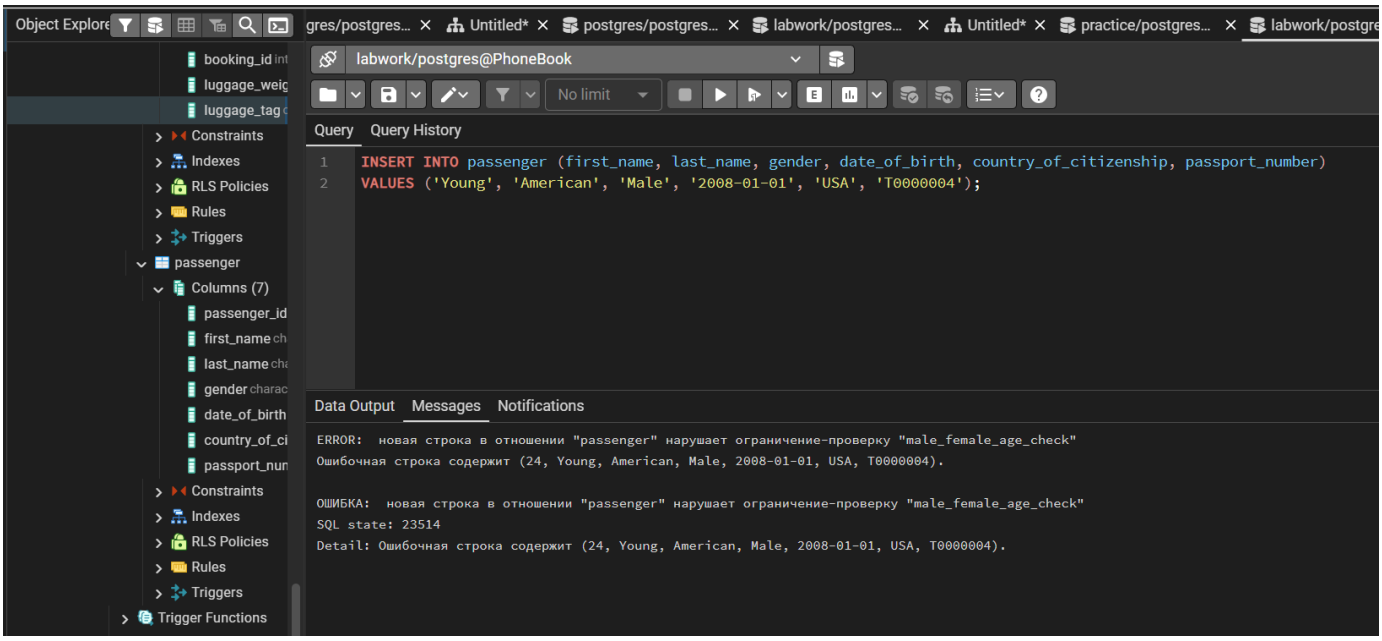
Task 7: Add a CHECK constraint to add rule as follow (use column country_of_citizenship):

- Passengers from Kazakhstan must be at least 18 years old.

- Passengers from France must be at least 17 years old.
- Passengers from other countries must be at least 19 years old.

ALTER TABLE passenger ADD CONSTRAINT pass_age_rest
CHECK(
(country_of_citizenship = 'Kazakhstan' AND date_of_birth <= CURRENT_DATE- INTERVAL '18 years') OR
(country_of_citizenship = 'France' AND date_of_birth <= CURRENT_DATE- INTERVAL '17 years') OR
(date_of_birth <= CURRENT_DATE- INTERVAL '19 years')
)



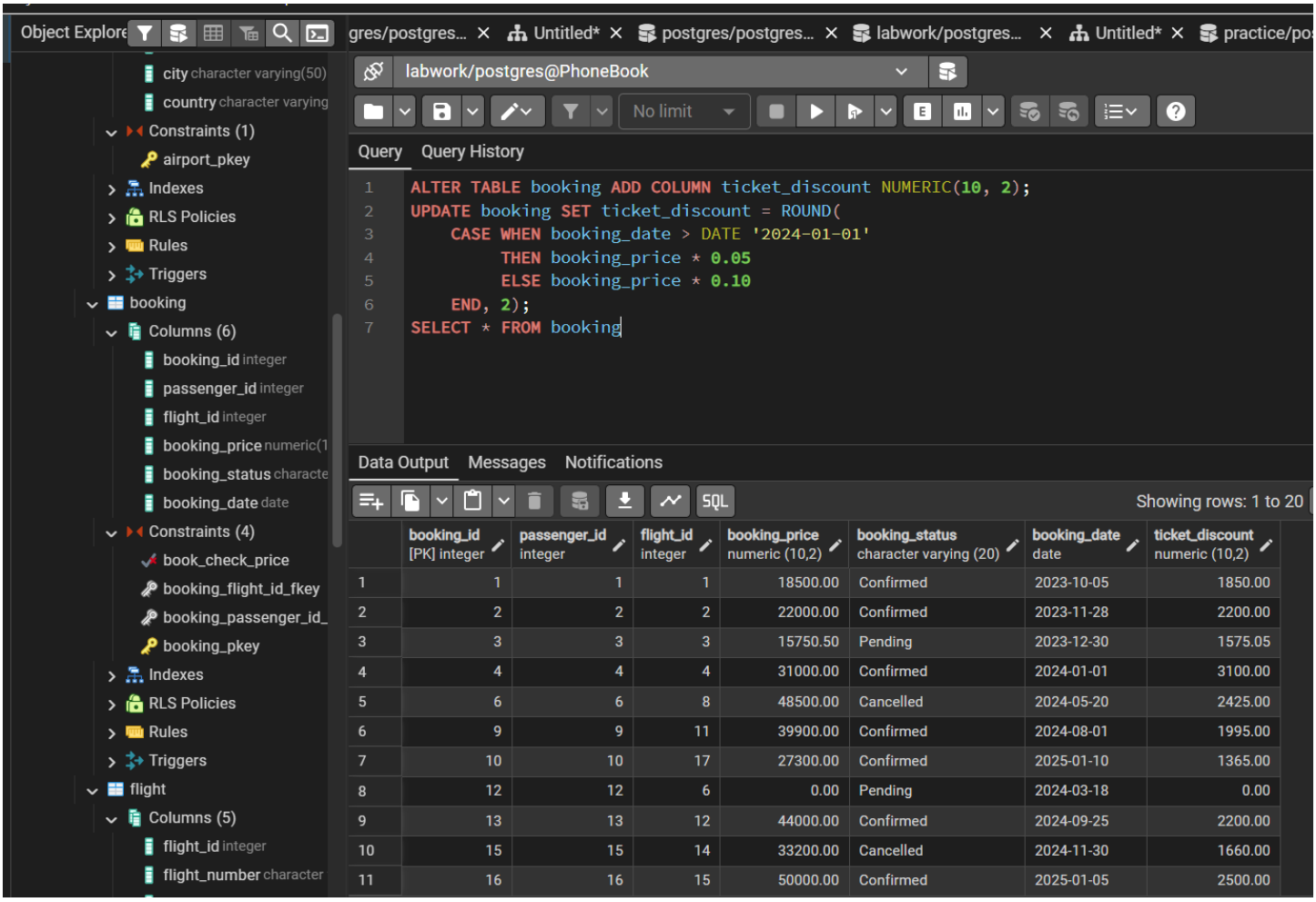


Task 8: Add a ticket_discount column to table booking and a CHECK constraint to apply some discount based on ticket price and created time:

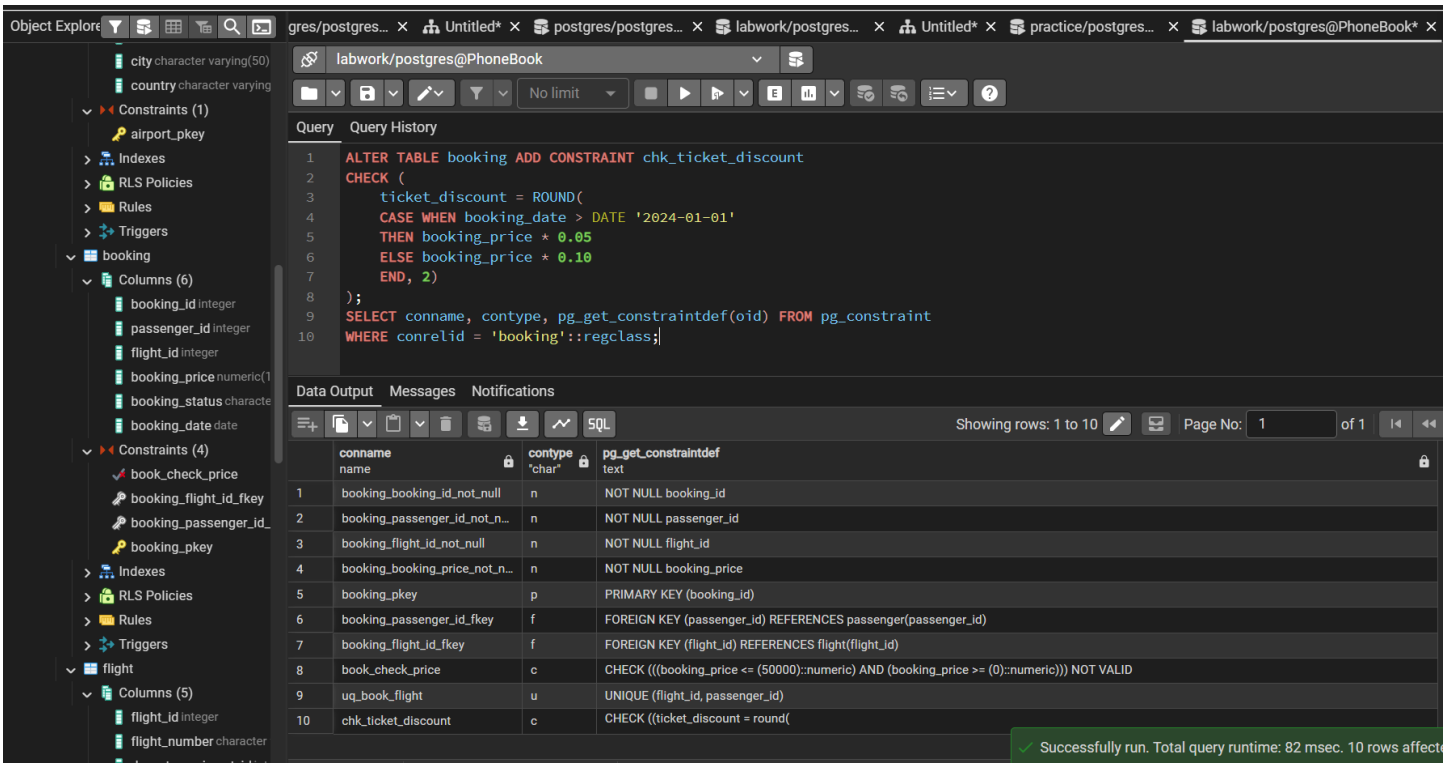
- the constraint applies a 5% discount for tickets created after 2024-01-01, and 10% discount for tickets created before 2024-01-01.

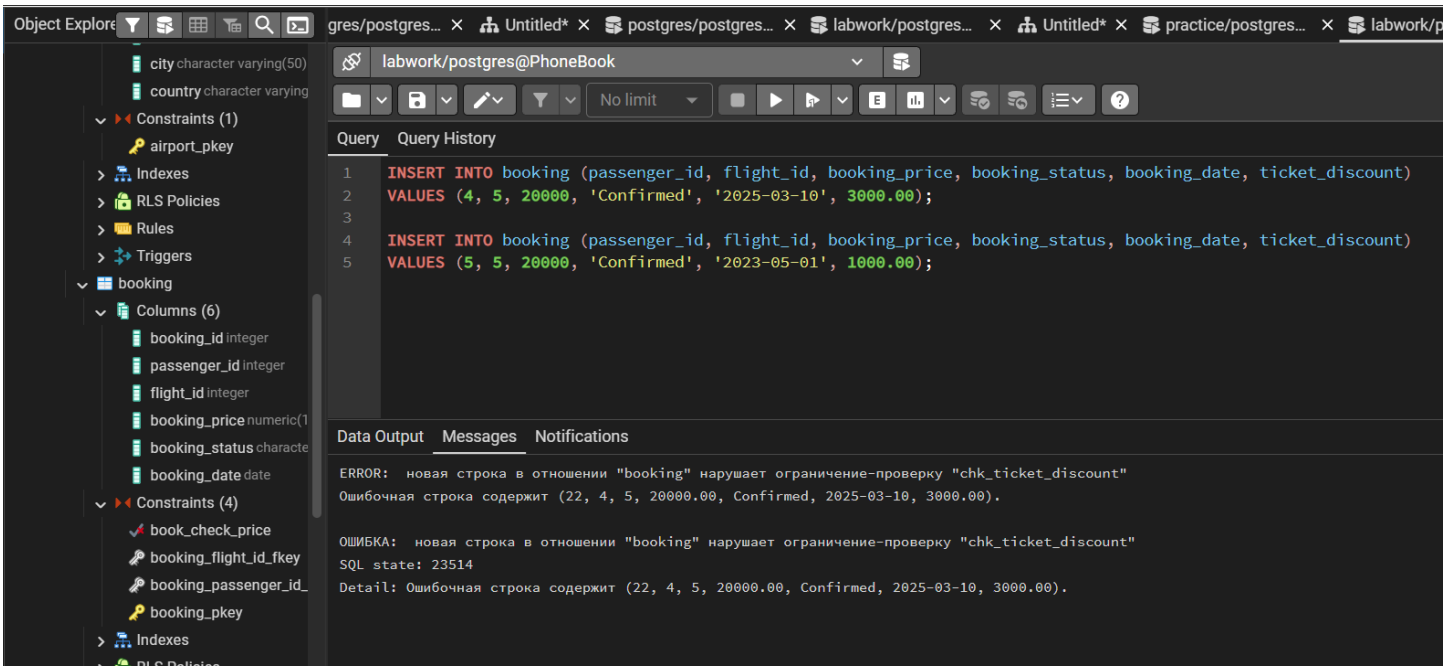
```
ALTER TABLE booking ADD COLUMN ticket_discount NUMERIC(10,2);

-- сначала заполняем существующие строки, иначе CHECK не добавится
UPDATE booking SET ticket_discount = ROUND(
   CASE WHEN booking_date > DATE '2024-01-01'
       THEN booking_price * 0.05
       ELSE booking_price * 0.10
   END, 2);
```




```
ALTER TABLE booking ADD CONSTRAINT chk_ticket_discount
CHECK (
   ticket_discount = ROUND(
      CASE WHEN booking_date > DATE '2024-01-01'
          THEN booking_price * 0.05
          ELSE booking_price * 0.10
      END, 2)
);
```



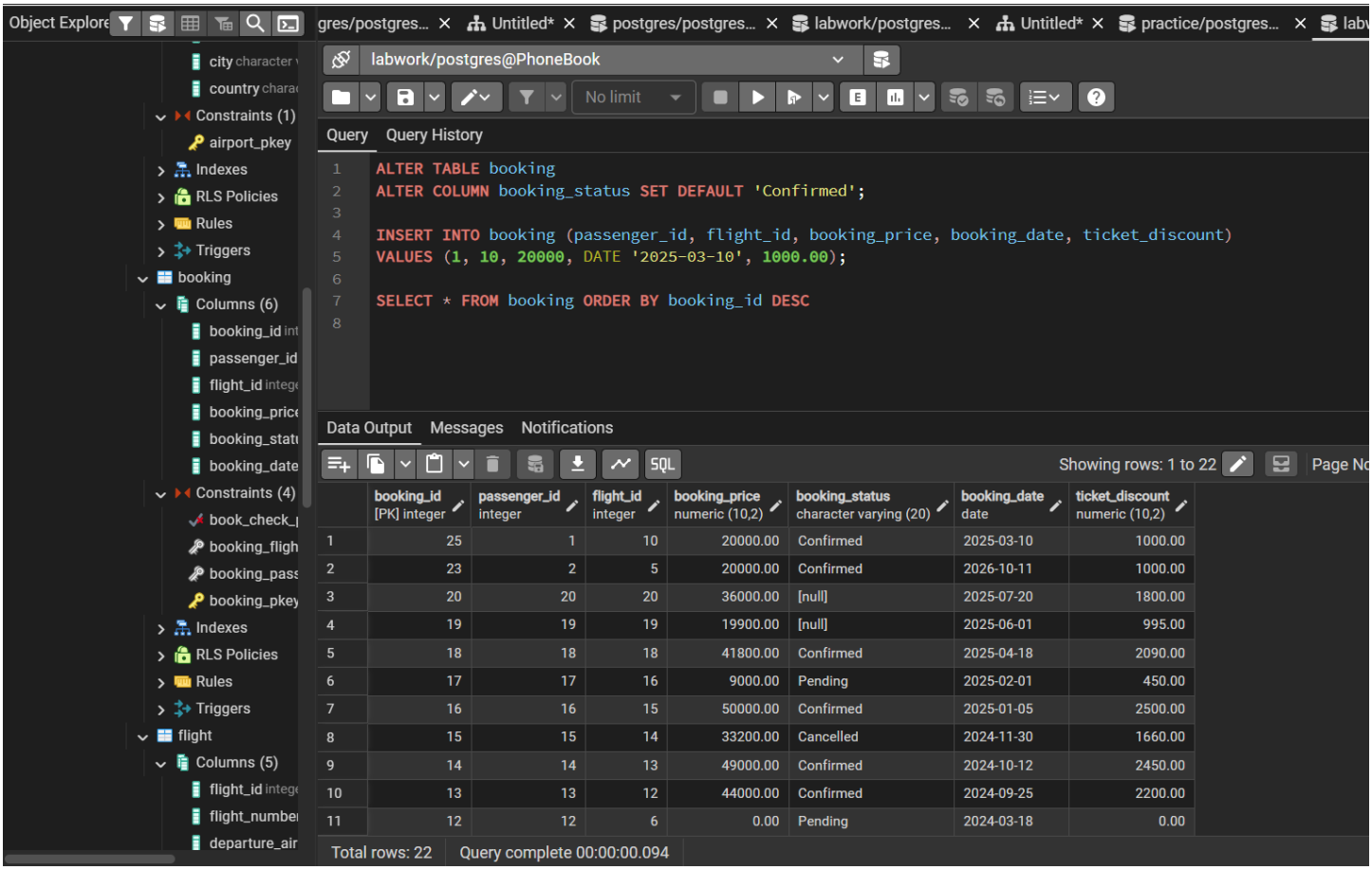


Task 9: Modify the booking table so that the booking status column has a default value of 'Confirmed'. Then insert a new booking without specifying booking status.

ALTER TABLE booking
ALTER COLUMN booking_status SET DEFAULT 'Confirmed';

INSERT INTO booking (passenger_id, flight_id, booking_price, booking_date, ticket_discount)
VALUES (1, 1, 20000, DATE '2025-03-10', 1000.00);

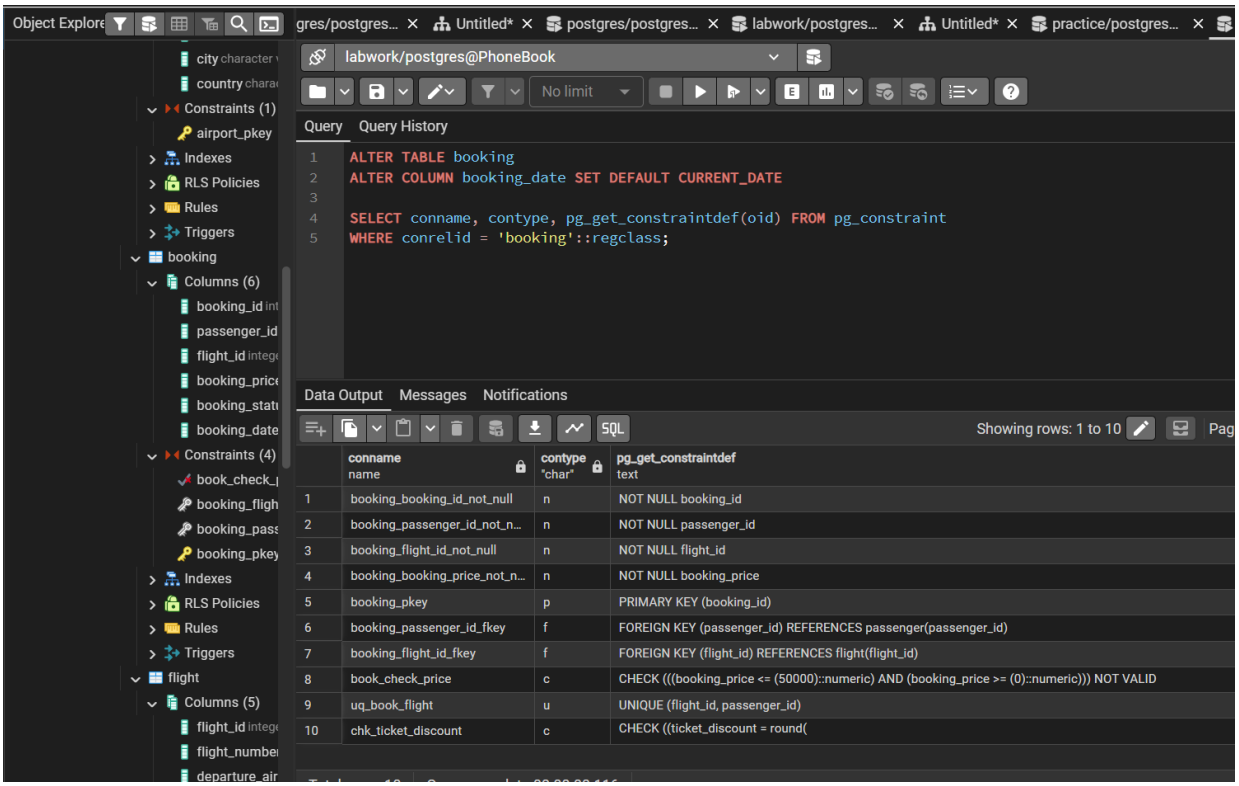
SELECT * FROM booking ORDER BY booking_id DESC LIMIT 1

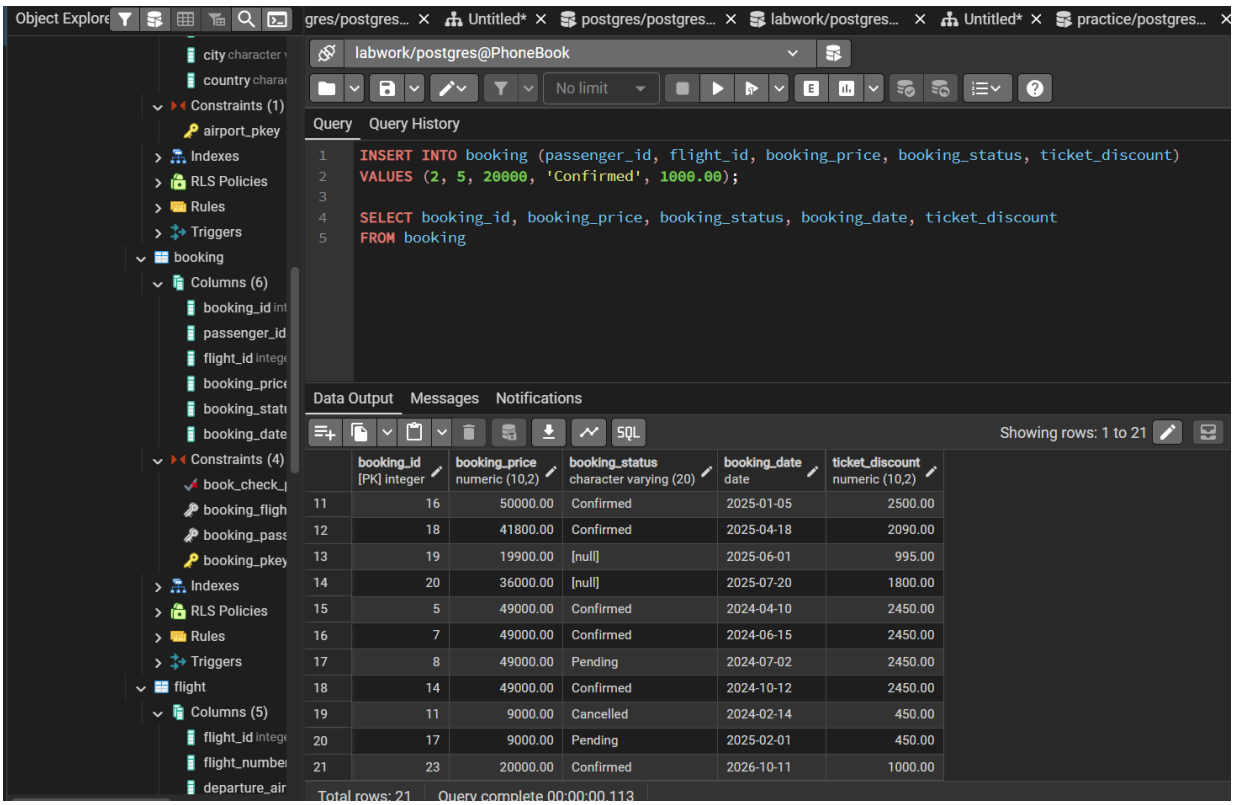


	booking_id [PK] integer	passenger_id integer	flight_id integer	booking_price numeric (10,2)	booking_status character varying (20)	booking_date date	ticket_discount numeric (10,2)
1	25	1	10	20000.00	Confirmed	2025-03-10	1000.00
2	23	2	5	20000.00	Confirmed	2026-10-11	1000.00
3	20	20	20	36000.00	[null]	2025-07-20	1800.00
4	19	19	19	19900.00	[null]	2025-06-01	995.00
5	18	18	18	41800.00	Confirmed	2025-04-18	2090.00
6	17	17	16	9000.00	Pending	2025-02-01	450.00
7	16	16	15	50000.00	Confirmed	2025-01-05	2500.00
8	15	15	14	33200.00	Cancelled	2024-11-30	1660.00
9	14	14	13	49000.00	Confirmed	2024-10-12	2450.00
10	13	13	12	44000.00	Confirmed	2024-09-25	2200.00
11	12	12	6	0.00	Pending	2024-03-18	0.00

Task 10: Modify the appropriate date column in the database so that when a new record is inserted, the date is automatically set to the current date.

ALTER TABLE booking
ALTER COLUMN booking_date SET DEFAULT CURRENT_DATE


Object Explorer
gres/postgres...
Untitled*
postgres/postgres...
labwork/postgres...
Untitled*
practice/postgres...
labwork/postgres@PhoneBook
No limit
Query
Query History
ALTER TABLE booking
ALTER COLUMN booking_date SET DEFAULT CURRENT_DATE
SELECT conname, contype, pg_get_constraintdef(oid) FROM pg_constraint
WHERE conrelid = 'booking'::regclass;
Data Output
Messages
Notifications
Showing rows: 1 to 10
conname name
contype "char"
pg_get_constraintdef text
1 booking_booking_id_not_null n NOT NULL booking_id
2 booking_passenger_id_not_n... n NOT NULL passenger_id
3 booking_flight_id_not_null n NOT NULL flight_id
4 booking_booking_price_not_n... n NOT NULL booking_price
5 booking_pkey p PRIMARY KEY (booking_id)
6 booking_passenger_id_fkey f FOREIGN KEY (passenger_id) REFERENCES passenger(passenger_id)
7 booking_flight_id_fkey f FOREIGN KEY (flight_id) REFERENCES flight(flight_id)
8 book_check_price c CHECK (((booking_price <= (50000)::numeric) AND (booking_price >= (0)::numeric))) NOT VALID
9 uq_book_flight u UNIQUE (flight_id, passenger_id)
10 chk_ticket_discount c CHECK ((ticket_discount = round(


Object Explorer
gres/postgres...
Untitled*
postgres/postgres...
labwork/postgres...
Untitled*
practice/postgres...
labwork/postgres@PhoneBook
No limit
Query
Query History
INSERT INTO booking (passenger_id, flight_id, booking_price, booking_status, ticket_discount)
VALUES (2, 5, 20000, 'Confirmed', 1000.00);
SELECT booking_id, booking_price, booking_status, booking_date, ticket_discount
FROM booking
Data Output
Messages
Notifications
Showing rows: 1 to 21
booking_id [PK] integer
booking_price numeric (10,2)
booking_status character varying (20)
booking_date date
ticket_discount numeric (10,2)
11 16 50000.00 Confirmed 2025-01-05 2500.00
12 18 41800.00 Confirmed 2025-04-18 2090.00
13 19 19900.00 [null] 2025-06-01 995.00
14 20 36000.00 [null] 2025-07-20 1800.00
15 5 49000.00 Confirmed 2024-04-10 2450.00
16 7 49000.00 Confirmed 2024-06-15 2450.00
17 8 49000.00 Pending 2024-07-02 2450.00
18 14 49000.00 Confirmed 2024-10-12 2450.00
19 11 9000.00 Cancelled 2024-02-14 450.00
20 17 9000.00 Pending 2025-02-01 450.00
21 23 20000.00 Confirmed 2026-10-11 1000.00
Total rows: 21
Query complete 00:00:00.113